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Comparison of traditional vs. lighter load strength training on fat-free mass, strength, power and affective responses in middle and older-aged adults: A pilot randomized trial

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ABSTRACT

Objectives: Strength training is widely recommended to improve strength, muscle mass and power. However, the feasibility and potential efficacy of strength training using lighter loads near failure on these outcomes in middle and older-aged adults remains unclear.

Methods: 23 community-living adults were randomized into two groups: Traditional strength training (ST) (8–12 repetitions) or a lighter load, higher repetitions (LLHR) (20–24 repetitions) group. Participants performed a full-body workout (twice a week) with 8 exercises at a perceived exertion of 7–8 (0–10 scale) for 10 weeks. Post-testing was performed by an assessor blinded to group assignments. An analysis of covariance (ANCOVA) was used to examine between group differences using baseline values as a covariate.

Results: The study involved individuals with a mean age of 59 years, of which 61 % were women. The LLHR group demonstrated a high attendance rate of 92 % (9.5 %) and reported leg press exercise RPE of 7.1 (0.53), along with a session feeling scale of 2.0 (1.7). There was a trivial difference in fat free mass (FFM) favoring LLHR vs ST [0.27 kg 95 % CI (−0.87, 1.42)]. The ST group exhibited superior increases in leg press 1 repetition maximum (1RM) strength [−14 kg (−23, −5)], while the LLHR group showed greater strength endurance increases (65 % 1RM) [8 repetitions (2, 14)]. Leg press power [41 W (−42, 124)] and exercise efficacy [−3.8 (−21.2, 13.5)] demonstrated trivial between-group differences.

Conclusion: A pragmatic, full-body strength training program with lighter loads taken close to failure appears to be a viable option for promoting muscular adaptations in middle- and older-aged adults. These results are exploratory and require a larger trial for confirmation.

1. Introduction

Maintaining muscle mass and strength is critical for preventing mobility disability and falls, and to maintain long-term independence in aging adults (Fried and Guralnik, 1997; Janssen et al., 2002; Bergen et al., 2016). It is now well-established that there is a progressive loss of muscle mass and strength with aging, termed as Sarcopenia (Cruz-Jentoft et al., 2019), which has been recognized as a muscle disease with an ICD-10 Diagnosis Code (Cao and Morley, 2016). Considering that the world aging population (65 years and over) is projected to double in number to approximately 1.5 billion by 2050, identifying effective interventions to maintain physical function is a significant public health challenge that demands immediate attention (United Nations Department of Economic and Social Affairs, 2020).

Physical activity is a major strategy to improve physical function in older adults, especially strength training (Zapata-Lamana et al., 2021; Parra-Rizo et al., 2022; Pahor et al., 2014; Balachandran et al., 2020). Strength training is widely recommended to increase muscle mass, strength, and physical function. Load or intensity, defined as the amount of weight lifted, is considered a crucial component of strength training.

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As such, the American College of Sports Medicine and the Physical Activity Guidelines for Americans recommend that older adults perform 8–12 repetitions to muscle fatigue, but not exhaustion (Piercy et al., 2018; American College of Sports Medicine et al., 2009). ACSM recommends using a load equivalent to 60–80 % of 1RM for 8–12 repetitions. Despite the compelling benefits and global recommendations for ST in older adults, fewer than 15 % of older adults (≥65 years) participate in muscle-strengthening activities in the US (Katzmarzyk et al., 2019). While highly effective, using higher loads/intensity can be challenging and burdensome for older adults, with potentially higher risk of injury and pain, particularly for those with joint-related issues. Several studies have identified common barriers to strength training among older adults, including fear of injury, pain and lack of access to exercise facilities (Burton et al., 2017a; Burton et al., 2017b). Notably, both younger and older individuals who engage in strength training tend to choose loads significantly lower than the current recommended amount, highlighting the difficulties associated with using heavy loads (Elsangedy et al., 2013; Steele et al., 2022).

Interestingly, studies in younger adults, both trained and untrained, have shown that lighter loads taken to failure are as effective as higher loads in increasing muscle growth (Schoenfeld et al., 2015; Morton et al., 2016; Schoenfeld et al., 2014; Schoenfeld et al., 2017). However, evidence supporting the effectiveness of lighter weights is lacking in older adults. Meta-analyses comparing high loads versus low-moderate loads in older adults show high loads to be marginally superior for muscle growth (Csapo and Alegre, 2016; Borde et al., 2015). This age discrepancy could be partly due to studies in older adults training with suboptimal effort when using lighter weights. Further, mechanistic studies have shown that training close to failure is necessary when using lighter weights to increase motor unit recruitment, particularly of the high-threshold muscle fibers, and thus maximize protein synthesis and muscle growth (Burd et al., 2010; Jenkins et al., 2015). Therefore, older adults who train with lighter weights may need to train to, or close to, failure to achieve the same benefits as higher loads (Dankel et al., 2017).

Due to the paucity of evidence on the topic, we conducted a pilot RCT by randomizing 23 middle-aged and older adults to a traditional strength training (8–12 repetitions) group (ST) or a lighter load, high repetitions (20–24 repetitions) group (LLHR) for 10 weeks to assess feasibility of the protocol. We also investigated the effects on fat-free mass, strength, power, physical function, and affective responses as secondary outcomes.

2. Materials & methods

2.1. Study design

The study employed a 10-week pilot randomized, controlled, single-blind design to assess body composition and function in independently living middle-aged and older persons. A CONSORT flow diagram of the study is presented in Fig. 1.

2.2. Sample size

Since this was a pilot study, a formal sample size calculation was not performed. Based on our limited resources, we estimated 12 per group would be large enough to assess feasibility and potential efficacy goals.

2.3. Participants

The participants were recruited through various methods such as flyers, posters, and advertisements in newsletters in the local New York Community. To be eligible, volunteers had to be between the ages of 50 and 80, live independently in the community, and be able to read and communicate in English. Exclusion criteria included neurological impairments, severe musculoskeletal impairment, major depression, severe vestibular problems, simultaneous use of cardiovascular, psychotropic and antidepressant drugs, participation in a structured exercise program using weights in the past six months, and active participation in a resistance or balance training program.

2.4. Randomization & allocation concealment

Randomization was stratified based on sex using a random block size of 2, 4, or 6 and generated through a web-based randomization system (www.sealedenvelope.com). To conceal allocation, cards were wrapped in foil to prevent transillumination. Subsequently, opaque envelopes

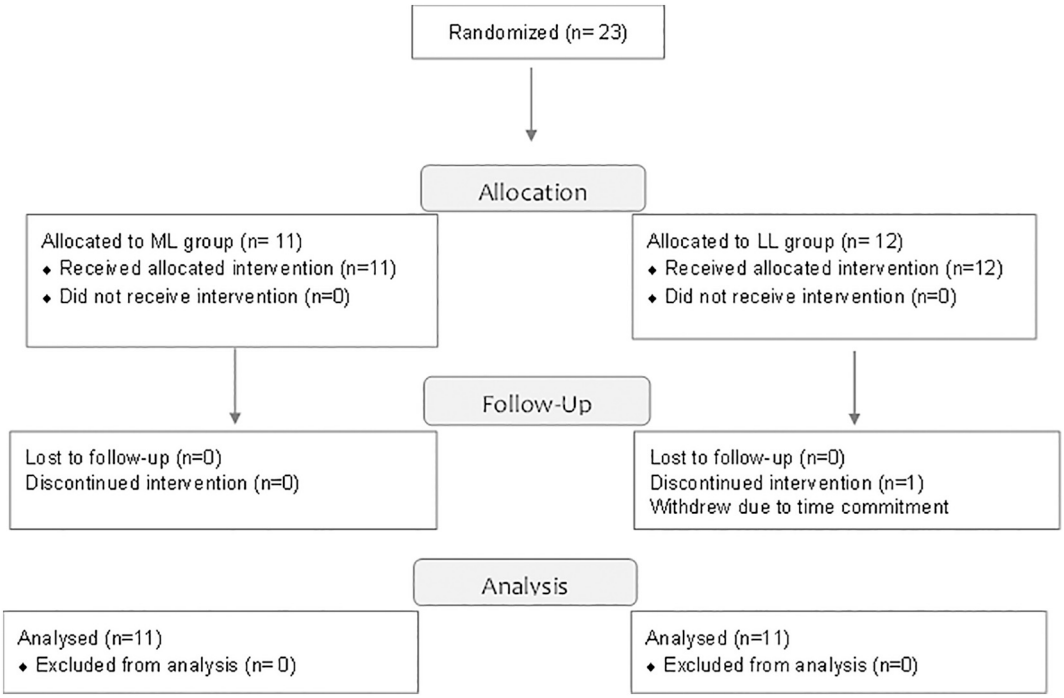

Fig. 1. Participant flow diagram.

were sequentially numbered, signed, and sealed by personnel not involved in the recruitment or assessment of participants.

2.5. Intervention

After randomization and prior to the start of training, training loads were determined by using a rating of perceived exertion (RPE) scale to maintain a target RPE of 7–8, where 10 is defined as the maximum perceivable effort (Steele et al., 2016). All participants were familiarized with the RPE scale and had to confirm their understanding before using it. Additionally, participants used the scale during the 1RM testing, which helped them understand the anchor points of 0 and 10.

Participants engaged in a 10-week, full-body workout twice per week. Within the 10-week training period, participants went through a 2- week preparatory phase during which the volume and intensity gradually increased to 2–3 sets and 7–8 RPE. Since the population was primarily untrained, this 2-week period was also used to assess and correct proper lifting techniques. Each training session began with 1 set of warm-ups at half the set workload for 8–10 repetitions on two multi-joint upper body and lower body machines.

The volume and progression of training were based on the ACSM resistance training guidelines, with participants maintaining an exercise intensity of 7–8 on an RPE scale of 0–10 and a volume of 2–3 sets. Participants were instructed to lift weights under control, and load was increased by 5–10 % when they could perform 2–3 sets of 12 or 24 repetitions while reporting an RPE of <7. A recovery period of 2–3 min was provided between sets for multi-joint exercises and 1–2 min for single-joint exercises. Both groups performed three lower body (leg press, leg extension and leg curls) and five upper body (chest press, shoulder press, seated rows, arm curls, and triceps pushdowns) exercises.

Participants performed all the exercises using conventional plate loaded resistance machines (Precor, Inc., Medway, MA, USA). Perceived exertion (RPE) was recorded for each participant after each exercise to track weight progressions and prior to cool down for session RPE. Each session ended with a 3–5 min cool down involving stretching exercises and light calisthenics. The training sessions in groups were supervised by at least two trainers who were exercise science majors.

2.6. Outcomes

Participants were assessed at baseline and at week 11 in the following order:

- Body composition: Fat-free mass and fat mass were obtained using a multifrequency bioelectrical impedance analyzer InBody770 (InBody USA, Cerritos, CA). The InBody 770 has shown a very high ICC for assessing fat-free mass (FFM) and fat mass compared to DEXA (Schoenfeld et al., 2020; McLester et al., 2020). The test was performed as per the manufacturer's specifications: tests were conducted in the morning, participants abstained from food and drink for at least 2 h prior to the test, no physical activity was performed 24 h prior to the test, and participants were instructed to empty their bladder prior to testing.
- Strength & Power: Lower body power was assessed using a pneumatic leg press (Keiser A300, Keiser Sports Health Equipment, Fresno, CA), as previously described (Balachandran et al., 2021). The seat position was adjusted to maintain 90° knee flexion at the starting position, and the highest peak power across three repetitions, with 1 min of rest between sets, was recorded as previously described (Balachandran et al., 2021). After assessment of power, lower body maximal strength was assessed by 1 repetition maximum (1RM) testing on the pneumatic leg press machine using ACSM guidelines (American College of Sports Medicine, 2021). After 3 sets of warm-ups with 50–70 % of their perceived maximum capacity, 1RM was obtained within 4–6 sets. A 2–3-min rest was provided between successive attempts and resistance was increased progressively by 2.5 to 20 kg. The criterion for stoppage was failure to complete a full repetition using proper form or refusal to continue testing. Strength endurance was assessed by using 65 % 1RM performed to failure (Van Roie et al., 2013). The total number of repetitions was recorded. Upper body strength was measured using a Jammar hand grip dynamometer on the dominant hand as previously described (Balachandran et al., 2021). The highest value across 2 attempts was recorded.
- Physical function: Physical function was assessed using the chair stand test performed as quickly as possible. The participants were instructed to sit in the middle of the chair, with their arms folded across their chest, and complete 10 chair stands as fast as possible as previously described (Rikli and Jones, 2013). The total time was recorded. The Get Up and Go (GUG) test involves standing from a chair, walking around a cone 3 m away, and sitting back down (Rikli and Jones, 1999). There were two trials per pace with 1 min rest, and the faster time was recorded.
- Patient-Reported Outcome Measures (PROMs): We used the Patient-Reported Outcomes Measurement Information System (PROMIS) physical function and mobility questionnaire developed by the National Institutes of Health (NIH) to evaluate self-reported measures of function (Fries et al., 2011; Fries et al., 2014). Participants used the PROMIS iPad App to complete the questionnaires, and we used the T-score for the final analysis.
- Affective responses: Participants rated their affective valence (pleasure/displeasure) using the Feeling Scale (FS) immediately after two exercises and after the exercise session (Hardy and Rejeski, 1989). The FS uses an 11-point bipolar scale, which ranges from very bad (−5) to very good (+5). The bipolarity of the FS conceptually maps on to the affective valence dimension of the circumplex model of affect (Russell, 1980). During the final workout week, participants completed the 18-item Physical Activity Enjoyment Scale (PACES) to assess physical activity enjoyment (Kendzierski and DeCarlo, 1991). The 18-item questionnaire showed a high internal consistency with a Cronbach's alpha of 0.92.
- Exercise Self-Efficacy: Participants also completed the exercise self-efficacy scale to assess their beliefs in their ability to continue strength training two times per week at the prescribed intensity in the future (McAuley, 1993).

2.7. Statistical analysis

Continuous variables were expressed as mean (SD) and categorical variables were as frequencies and percentages. An analysis of covariance (ANCOVA) was used to examine between group differences using baseline values as a covariate. For outcomes without a pretest, we used an independent *t*-test. All results are presented with their corresponding point estimates and 95 % confidence intervals (CI). Additionally, we used the means and pooled SD as standardizers to calculate Hedge's g. A Hedge's g effect size estimate was used because it includes a correction for sample size and is therefore more appropriate to use with small samples. The interpretation of g is similar to the interpretation of the Cohen d: 0.80 is considered large, 0.50 is considered medium, and 0.20 is considered small. Data were analyzed using both an intention-to-treat approach (ITT) and per protocol analysis (PP). All analyses were conducted in R (Version 4.1.3). Due to the exploratory nature of the pilot study, we did not perform any hypothesis testing.

3. Results

As shown in Fig. 1, 23 participants were recruited and randomized into two groups. Baseline characteristics were similar between the groups (see Table 1). The mean participant age was 58.5 years and 61 % were female.

Table 1
Baseline participant characteristics.

	ST (N = 11)	LLHR (N = 12)
Age, yrs.	58 ± 5	58 ± 4
Male	4 (36 %)	5 (42 %)
Body mass index	29 ± 9	26 ± 3
Race/ethnicity		
White	9 (82 %)	6 (50 %)
African American/Black	0 (0 %)	3 (25 %)
College education	7 (64 %)	7 (58 %)
Body composition		
Fat free mass, kg	52.5 ± 13.9	50.1 ± 10.1
Body fat %, kg	34.4 ± 12.0	32.5 ± 9.3
Strength & power		
1RM, kg	146 ± 57	157 ± 54
65 % 1RM, kg	95 ± 37	101 ± 36
Leg press power, W	1076 ± 427	1152 ± 463
Grip strength, kg	38 ± 13.1	38.25 ± 11.2
Physical function		
Get up & go, s	5.7 ± 0.6	5.6 ± 0.5
Chair stands, s	20.8 ± 4.3	18.2 ± 4.1
Self-reported function		
PROMIS physical function	50.4 ± 7.6	50.3 ± 5.3
PROMIS mobility	51.2 ± 6.4	52.8 ± 7.5
Conditions		
Hypertension	3 (27 %)	3 (25 %)
Diabetes	1 (9 %)	0 (0 %)
Cholesterol	3 (27 %)	3 (25 %)

Abbreviations: 1RM, One Repetition Maximum; PROMIS; Patient Reported Outcomes Measurement Information System.

3.1. Intervention adherence

The LLHR group attended 92 % (SD, 9.5 %) and ST group attended 91 % (SD,6.4 %) of their sessions.

3.2. Body composition, strength, power & physical function

As shown in Table 2, the majority of outcomes showed trivial differences between groups and wide confidence intervals.

For body composition, FFM increased (ST: 0.24 kg, 1 %; LLHR: 0.67 kg, 1.4 %) and showed a trivial difference between groups 0.27 kg, 95 % CI (−0.87, 1.42). PP analysis was similar (ST: 0.52 kg, 1 %; LLHR: 0.85 kg, 2 %) with a trivial between-group difference 0.22 kg, 95 % CI (−0.87, 1.31).

For leg press strength (1RM), both groups improved strength (ST: 20 kg, 14 %; LLHR: 8 kg, 5 %) and showed a small effect −14 kg (−23, −5) in favor of the ST group. Strength endurance (repetitions) (ST: 2 reps, 6.1 %; LLHR: 10 reps, 37 %), however, showed a moderate effect 8 repetitions (2, 14) in favor of the LLHR group. Leg press power (ST: 86.60 W, 8 %; LLHR: 110 W, 9.6 %) showed a moderate effect 41 (−42, 124) in favor of the LLHR group.

For physical function, chair stands (ST: −4.3 s, 21 %; LLHR: −2.2 s, 12 %) showed a small effect 1.66 s (0.44, 2.88) favoring the ST group, while GUG (ST: −0.6 s, 11 %; LLHR: −0.7 s, 13 %) showed a small effect −0.12 s (−0.53, 0.29) favoring the LLHR group. For self-reported function, PROMIS mobility 2.29 (−2.96, 7.53) and physical function 1.95 (−3.23, 7.13) showed a small and trivial effect in favor of the LLHR group.

The per-protocol analysis reported in the supplementary Table S1 showed similar results as the main analysis.

3.3. Affective responses

As shown in Table S3, the session FS scores for ST 2.4 (1.8) and LLHR 2.0 (1.7) ranged between the fairly good (1) and good (3) categories. Exercise FS scores reported immediately after chest press exercise were ST 2.3 (1.9) and LLHR 1.6 (1.9), and after leg press exercise were ST 2.4 (1.9) and LLHR 1.7 (1.7). The Physical Activity Enjoyment Scale (PACES) score − 14.9 (−32.7, 2.9) showed a moderate effect, and exercise efficacy −3.8 (−21.2, 13.5) showed little difference between groups.

Perceived session exertion was ST 7.0 (0.76) and LLHR 6.9 (0.60). Perceived exertion for chest press exercise ST 7.1, (0.89) & LLHR 7.3 (0.98) and leg press exercise ST (6.7 (0.85)) & LLHR 7.1, (0.53). FS &

Table 2
Raw, adjusted means, and effect size for all outcomes.

	Mean at 10 wks (SD) (Post group)		Adjusted mean at 10 wks (SE)		Adjusted mean difference (95 % CI)	Effect size Hedges' g 95 % CI
	ST	LLHR	ST	LLHR	LLHR - ST	LLHR - ST
Body composition						
Fat free mass, kg	52.7 ± 13.1	50.2 ± 10.2	51.3 ± 0.4	51.6 ± 0.4	0.27 (−0.87, 1.42)	0.02 (−0.32, 0.36)
Body fat %	34.3 ± 11.9	32.3 ± 9.9	33.6 ± 0.5	33.0 ± 0.5	−0.69 (−2.18, 0.81)	−0.06 (−0.40, 0.28)
Body fat, kg	29.9 ± 16.2	24.2 ± 8.6	27.4 ± 0.6	26.8 ± 0.6	−0.63 (−2.28, 1.03)	−0.05 (−0.39, 0.29)
Strength & power						
1RM, kg	166 ± 63	168 ± 62	173 ± 3	160 ± 3	−14 (−23, −5)	−0.21 (−0.55, 0.13)
65 % 1RM, reps	28 ± 11	35 ± 12	28 ± 2	36 ± 2	8 (2, 14)	0.67 (0.31, 1.04)
Grip strength, kg	39.0 ± 14.0	39.2 ± 11.8	39.0 ± 1.1	39.2 ± 1.1	0.27 (−3.13, 3.68)	0.02 (−0.32, 0.36)
Leg press power, w	1153 ± 397	1284 ± 497	1198 ± 28	1239 ± 28	41 (−42, 124)	0.09 (−0.25, 0.43)
Physical function						
GUG, s	5.1 ± 0.6	4.8 ± 0.6	5.0 ± 0.1	4.9 ± 0.1	−0.12 (−0.53, 0.29)	−0.19 (−0.53, 0.15)
Chair stands, s	16.5 ± 3.6	15.4 ± 3.4	15.1 ± 0.4	16.8 ± 0.4	1.66 (0.44, 2.88)	0.45 (0.10, 0.80)
Self-reported						
PROMIS Function	53.3 ± 7.9	54.9 ± 6.5	53.1 ± 1.8	55.1 ± 1.8	1.95 (−3.23, 7.13)	0.26 (−0.08, 0.61)
PROMIS Mobility	52.5 ± 7.1	56.0 ± 6.8	53.1 ± 1.8	55.4 ± 1.8	2.29 (−2.96, 7.53)	0.32 (−0.03, 0.66)
PACES*	111 (16)	96.3 (20.7)			−14.9 (−32.7,2.9)	−0.76 (−1.7, 0.09)
EXSE*	88.8 ± 17.3	85.0 ± 19.74			−3.8 (−21.2, 13.5)	−0.19 (−1.0, 0.64)

Abbreviations: 1RM, One Repetition Maximum; STS peak power, Sit to Stand peak power; GUG, Get up & Go; PROMIS, Patient Reported Outcomes Measurement Information System PACES, Physical Activity Enjoyment Scale; EXSE, Exercise Self-Efficacy Scale. *No pre-testing.

RPE scores for all participants are shown in supplementary Fig. S1.

3.4. Adverse events

There were no serious adverse events reported, with most of the events reported as mild/moderate severity as shown in Table 3. There were four events in the ST group and one in the LLHR group that were categorized as "possibly related" to the intervention. Musculoskeletal events were the most common (shoulder, arm, back, and knee pain).

3.5. Drop-outs

One female participant dropped out from the LLHR group after four sessions due to time commitment.

4. Discussion

This pilot study compared light load, high repetition training to traditional ST for 10 weeks in 23 middle and older-aged adults. The study showed that a pragmatic, full-body ST with lighter loads (20–24 repetitions) taken close to failure is feasible, acceptable, and shows potential efficacy for enhancing muscular adaptations. Both groups improved FFM, strength, power, and physical function and strength endurance.

Feasibility assessed using attendance (>90 %) and compliance to the protocol (exercise RPE scores), showed that lighter loads taken to volitional fatigue is feasible in middle and older aged adults. Moreover, all the participants adhered to the protocol, indicating that the protocol was perceived as acceptable. Retention was high (92 %) with only one participant dropping out from the LLHR group due to lack of time.

To our knowledge, only one study specifically compared training to failure with high loads (80 % 1RM) vs light loads (20 % 1RM) in healthy older adults (Van Roie et al., 2013). The study showed upper leg volume, measured using computed tomography (CT) scan, to be similar in both groups (3.2 % vs. 2.4 %, respectively). However, the training program was limited to two lower-body exercises performed for one set of 20 % 1RM for 80–100 repetitions to volitional failure, and progression was based on 1RM testing every 4 weeks, which questions the feasibility and generalizability of the training program. In contrast, our study employed a pragmatic protocol using a full-body routine with 20–24 repetitions performed near failure, and progressions were based on RPE and repetitions completed.

Testing 1RM may not be feasible or safe for untrained individuals without expert guidance. Multicenter physical activity trials routinely employ RPE to select and progress loads for strength training interventions (Pahor et al., 2014; Stathi et al., 2022). Furthermore, RPE vs 1RM testing shows little difference in outcomes in elderly (Buskard et al., 2019). We also chose a vigorous intensity RPE (7–8) instead of having the participants train to complete failure, which can be extremely challenging (Fisher and Steele, 2017) and may not be feasible in the long-term.

As shown in Table 1 and Fig. 2, both groups improved FFM with trivial between-group differences. A meta-analysis of 49 studies, including both randomized and non-randomized studies, in older adults showed an increase of approximately 1 kg in lean body mass with an average of 20.5 weeks (Peterson et al., 2011). The majority of strength training studies in older adults had a duration of 12 to 26 weeks, adopted a higher frequency (3/week), resulting in a lean body mass or fat-free mass change from 0 kg to 2 kg (Borde et al., 2015; Peterson et al., 2011; Balachandran et al., 2023). Overall, the magnitude of FFM increase for the short duration (8-week excluding the preparatory phase) program with a low frequency (2/week) is in accordance with other

Table 3
Adverse effects.

	ST	LLHR
	Participants (%)	Participants (%)
Serious events[a]	0	0
Definitely related[a]		
Mild	0	0
Moderate	0	0
Severe	0	0
Possibly related[a]		
Mild	4 (36.3)	1 (8.3)
Moderate	0	0
Severe	0	0
Not related[a]		
Mild	3 (27.2)	0
Moderate	0	3 (25)
Severe	1 (9)	0

[a] Classification definitions given in supplementary Table S2.

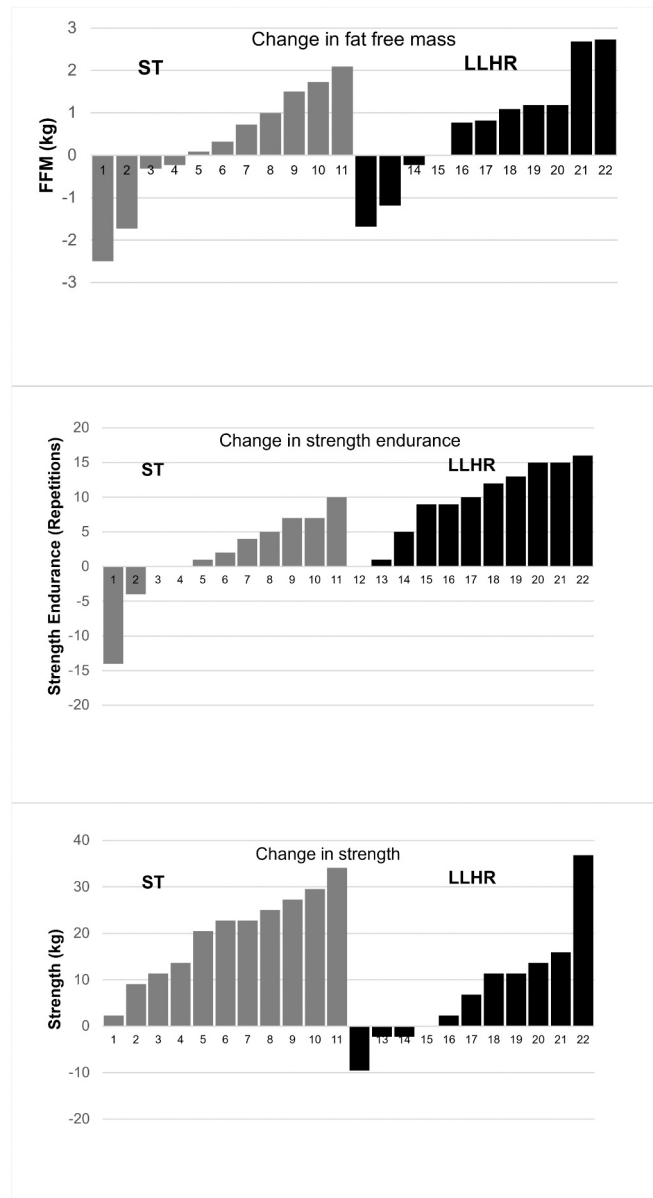

Fig. 2. Change in fat free mass, strength, and strength endurance in participants after 10 weeks.

strength training studies in older adults.

For strength, both groups improved from baseline, but the ST group showed almost a 14 kg, greater improvement compared to the LLHR group, with interval limits ranging from 5 kg to 23 kg. Strength endurance (reps performed at 65 % 1RM) increased from baseline in both groups; however, the LLHR group performed six repetitions more than the ST group with limits ranging from 2 to 13 repetitions. The study that compared 80 % Vs. 20 % 1RM (Van Roie et al., 2013) showed similar magnitude of improvement (46 % vs. 23 %) to ours for strength, but not for endurance (14.6 % vs. 19.8 %; 1.1 repetitions more). We believe this could be partly attributed to the difference in 1RM methodology used between studies: Van Roie et al. (2013) estimated leg extension 1RM using a regression equation whereas we directly assessed using a leg press 1RM test.

Peak muscle power, as assessed by the leg press test, showed improvements in both groups and trivial differences between groups. Physical function, measured using both the 10 chair stands and GUG test, showed improvements. Notably, the 10 chair stands showed a small 1.7 s improvement in the ST group compared to LLHR, with the range of improvement varying from 0.4 to 2.9 s. Chair stands require higher percentage of maximal strength to raise the body weight and thus could have favored the ST group, which showed large improvement in strength (Cuoco et al., 2004). However, the GUG test showed trivial differences. Akin to our study, the study comparing 80 % vs. 20 % 1RM (Van Roie et al., 2013) showed greater improvements (8.9 % vs. 5 %) in 30 s chair stands in the high group compared to the low load group, and trivial differences in the GUG test (2.9 % vs. 3.9 %). Self-reported function assessed using PROMIS questionnaires (physical function and mobility) improved in both groups.

For affective responses, participants reported a mean session score of +2 (between good and very good). Both session and exercise FS scores favored the ST group, with wide CI's as shown in Table S3. The lower scores for lighter loads could be attributed to the increase in physiological markers of anaerobic fatigue, leading to greater discomfort (Fisher and Steele, 2017). Following the pattern for FS, exercise enjoyment favored the ST group. However, exercise efficacy, the ability to continue exercising at the prescribed intensity, showed little difference between groups. Although not collected systematically, several participants were unwilling to progressively increase the weight used in the leg press for fear of potential injury or pain, especially in the ST group. Anecdotally, several ST participants expressed concerns over increasing weights. On the contrary, several LLHR participants commented on the discomfort associated with high repetitions.

4.1. Limitations & strength

Our study had several limitations: First, the study can be considered a pilot trial; thus, neither outcomes nor feasibility benchmarks were preregistered. Additionally, the study was not powered to detect statistically significant differences, so any between-group comparison should be interpreted with caution. Second, although we used BIA to assess FFM, DXA is considered the "gold standard" for assessing changes in lean mass and thus may be more sensitive to detecting subtle changes over time. However, given that multi-frequency BIA shows good correlation with DXA for assessing FFM during structured ST programs on a group level (Schoenfeld et al., 2020), the practical implications of differences between methods is questionable. Third, we used a mixed-age (50–80 years) group due to COVID restrictions. Fourth, the study had a short follow-up period (8 weeks), with an initial 2 weeks as a preparatory phase; thus, the long-term effects on physical and affective outcomes are unclear. Fifth, our intervention was carried out in a research setting under the supervision of trainers. The generalizability of the intervention to an unsupervised, home-based environment using bands and ankle weights is therefore unclear.

The study also had several strengths: First, we used a pragmatic training protocol that included a full-body program, used perceived exertion to select and progress load, and participants systematically performed exercises near failure rather than to failure. Second, we tracked affective responses, which could potentially play a role in long-term adherence and self-efficacy. Third, we employed a rigorous methodology by using concealed allocation, both an intention-to-treat and per-protocol approach, systematically recording adverse events, and blinding assessors for outcome assessment.

4.2. Conclusions

Our study showed that pragmatic, full-body high-intensity strength training with lighter loads (20–24 repetitions) to volitional failure (7–8 RPE) is feasible and acceptable in healthy middle- and older-age individuals. Furthermore, both groups improved FFM, strength, power, physical function and strength endurance. Considering the feasibility, acceptability and potential efficacy of lighter load training, a long-term definitive study with a larger sample is required to confirm these findings.

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CRediT authorship contribution statement

Study concept & design: Balachandran, Quiles, Schoenfeld, Zenko

.Acquisition, analysis, or interpretation of data: Balachandran, Szabo, Watts, Quiles, Schoenfeld, Zenko.

Manuscript drafting: Balachandran.

Critical revision of the manuscript for important intellectual content: All authors.

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Final approval of the version to be published: All authors.

Declaration of competing interest

Competing Interests: BJS serves on the scientific advisory board of Tonal Corporation, a manufacturer of fitness equipment.

Data availability

www.osf.io/evdmg/

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.exger.2023.112219.

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